

HW02

ENAE 601: Astrodynamics

September 30th. at 23:59

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Problem 1

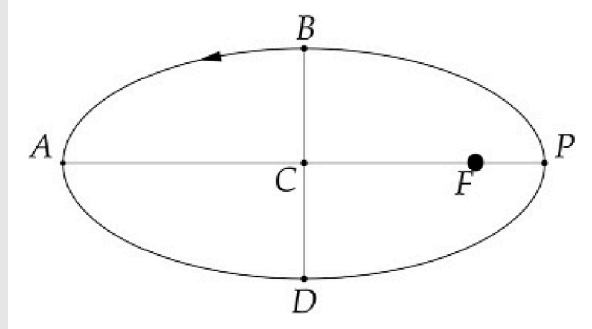
Where Curtis says “orbital elements”, he substitutes angular momentum for the standard semi-major axis. Show how to compute one from the other.

For the following problems and the rest of the semester, compute semi-major axis instead of angular momentum when classical (Keplerian) orbit elements are requested, unless otherwise specified.

Answer

Problem 2**Curtis 3.5**

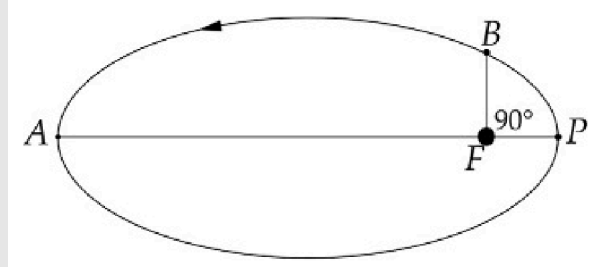
Calculate the time required to fly from P to B , in terms of the eccentricity e and the period T . B lies on the minor axis.



Answer

Problem 3**Curtis 3.6**

If the eccentricity of the elliptical orbit is 0.3, calculate, in terms of the period T , the time required to fly from P to B .



Answer

Problem 4**Curtis 3.8**

A satellite is in Earth orbit for which the perigee altitude is 200 km and the apogee altitude is 600 km. Find the time interval during which the satellite remains above an altitude of 400 km.

Answer

Problem 5**Curtis 3.9**

An Earth-orbiting satellite has a perigee radius of 7000 km and an apogee radius of 10000 km.

- a) What true anomaly $\Delta\theta$ is swept out between $t = 0.5$ h and $t = 1.5$ h after perigee passage?
- b) What area is swept out by the position vector during that time interval?

a) Answer

b) Answer

Problem 6**Curtis 3.10**

An Earth-orbiting satellite has a period of 14 h and a perigee radius of 10000 km. At time $t = 10$ h after perigee passage, determine:

- a) The radial position.
- b) The speed.
- c) The radial component of the velocity.

a) Answer

b) Answer

c) Answer

Problem 7**Curtis 3.15**

A spacecraft on a parabolic trajectory around the Earth has a perigee radius of 6600 km.

- a) How long does it take to coast from $\theta = -90^\circ$ to $\theta = +90^\circ$?
- b) How far is the spacecraft from the center of the Earth 36 h after passing through perigee?

a) Answer

b) Answer

Problem 8**Curtis 3.16**

A spacecraft on a hyperbolic trajectory around the Earth has a perigee radius of 6600 km and a perigee speed of $1.2v_{\text{esc}}$.

- a) How long does it take to coast from $\theta = -90^\circ$ to $\theta = +90^\circ$?
- b) How far is the spacecraft from the center of the Earth 24 h after passing through perigee?

a) Answer

b) Answer

Problem 9**Curtis 3.17**

A trajectory has a perigee velocity $1.1v_{\text{esc}}$ and a perigee altitude of 200 km. If at 10 a.m., the satellite is travelling towards the Earth with a speed of 8 km s^{-1} , how far will it be from the Earth's surface at 5 p.m. the same day?

a) Answer

Problem 10**Curtis 3.18**

An incoming object is sighted at an altitude of 100 000 km with a speed of 6 km s^{-1} and a flight path angle of -80° .

- a) Will it impact the Earth or fly by?
- b) What is the time to impact or to closest approach?

a) Answer

b) Answer

Code

```
./src/index.py Python
1  import numpy as np
2  import matplotlib.pyplot as plt
3  import os
4
5
6  def heading(problem):
7      print(f"\n-----\n {problem}\n-----")
8
9
10 def main():
11     #####
12     # p02 #
13     #####
14     heading("p02")
15
16     x = np.linspace(0, 10, 100)
17     y = np.sin(x)
18
19     print(f"x:\n{x}\nny:\n{y}")
20     with open("./outputs/text/s02.txt", "w", encoding="utf-8") as f:
21         f.write(f"x:\n{x}\nny:\n{y}")
22
23     plt.figure()
24     plt.plot(x, y)
25
26     save_path = os.path.join(os.path.dirname(__file__), "./outputs/figures/s02.png")
27     plt.savefig(save_path)
28     plt.show()
29
30 if __name__ == "__main__":
31     main()
```